

KESHAV KRISHNA

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Product engineer with three years shipping ML, NLP, and foundation-model systems to production, including consumer systems serving 100M+ users. Currently building risk analytics tooling at Moore Capital Management after extending a full-time summer internship into a part-time fall role. I own features from an ambiguous conversation through production, and I measure model quality with explicit evaluation and benchmarking frameworks.

EDUCATION

New York University, Courant Institute — MS, Computer Science, Artificial Intelligence <i>GPA 3.95/4.0; Machine Learning, Building LLM Reasoners, Honors Algorithms</i>	Aug 2025 – May 2027 New York City
Indian Institute of Technology, Ropar — BTech, Computer Science and Engineering <i>GPA 8.74/10; Merit Scholarship for top academic performance</i>	Aug 2019 – Jun 2023 Ropar, India

EXPERIENCE

Moore Capital Management — Risk Technology Intern <i>Risk analytics tooling and dashboards for internal users; full-time summer, part-time fall</i> - Owned risk tooling end to end: Python data workflows behind HTML/CSS dashboards, gone from verbal stakeholder request to production in days and now used daily by 20+ risk, portfolio, and operations staff. - Optimized VaR, CVaR, beta, and correlation calculations with vectorized Python and Numba JIT, cutting runtime about 70% on multi-million-row datasets so daily risk numbers are ready before market open. - Benchmarked distributed data-processing tools on the firm's compute cluster with profiling and system modeling, locating scalability bottlenecks that guided infrastructure decisions and lifting batch throughput about 2x.	May 2026 – Present New York City
JioSaavn — Software Engineer, Data Science <i>Production ML and NLP in a consumer app serving 100M+ users</i> - Built recommendation and personalization systems on embeddings and vector retrieval (Spark, Hive, Redis, Qdrant), driving 1M impressions, 800K clicks, and 500K streams per day in production. - Shipped a LLaMA3-8B-Instruct foundation-model system that combines intent understanding, retrieval-augmented generation, and automated song selection; owned deployment and inference optimization on Docker and Kubernetes, serving 50K+ generated playlists per day at sub-2s p95 latency. - Designed the benchmark dataset, evaluation metrics, and acceptance criteria for multilingual playlist moderation, then ran a bake-off across SetFit, HingRoBERTa, mURIL, and LLaMA3-8B-Instruct; the selected model hit 85% accuracy and absorbed about 60% of the manual review load. - Owned model, pipeline, and deployment for 3+ shipped systems over two years, working cross-functionally with product managers, domain experts, and platform engineering teams.	Jul 2023 – Jul 2025 Mumbai, India
GE Healthcare — Software Intern, Medical Computer Vision <i>Detection and de-identification pipelines for clinical imaging</i> Built YOLO-based medical-image de-identification and RGB/IR people-detection pipelines over 20K+ images, reaching 95% detection accuracy and cutting manual image review about 80%.	May 2022 – Jul 2022 Bangalore, India

SELECTED TECHNICAL PROJECTS

FlashAttention-2 — from-scratch Triton implementation [GitHub] - Implemented forward and backward kernels with online softmax, tiling, causal masking, and activation recomputation; cut attention memory from $O(N^2)$ to $O(N)$, taking 65K-token attention from 10+ GB to under 1 GB and making long-context training feasible. - Profiled Transformer models up to 2.7B parameters with NVIDIA Nsight Systems and PyTorch memory snapshots, benchmarking against vanilla PyTorch and torch.compile; measured about 2x speedup over naive attention.	2026
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Transformer language model from scratch [GitHub]

2026

- Built a 17M-parameter model in PyTorch from low-level components (BPE tokenization, RoPE, RMSNorm, SwiGLU, causal attention, AdamW, cosine LR scheduling), reaching validation loss **3.1**, with multiprocessing tokenization and memory-mapped loading cutting data-pipeline time about **3x**.
- Ran systematic ablations across pre- vs. post-norm, RoPE vs. NoPE, and SwiGLU vs. SiLU; the pre-norm + RoPE + SwiGLU recipe won, cutting validation loss about **8%** against the baseline configuration.

Pico-LLM — training, evaluation, and inference platform [GitHub]

2025

- Built a unified platform for SFT and DPO/GRPO training, Model Ops (checkpoint management, batch evaluation), and inference across Transformer, LSTM, and K-gram architectures.
- Added KV-cached generation and multi-GPU batch generation, lifting inference throughput about **3x** and improving experiment reliability across architectures.

Alignment and Reasoning RL for Mathematical LLMs [GitHub]

Spring 2026

- Built an end-to-end reasoning and post-training pipeline: zero-shot evaluation, SFT, GRPO with verified rewards, masked losses, entropy tracking, and policy-gradient updates.
- Fine-tuned Qwen2.5-Math-1.5B on chain-of-thought data through SFT and GRPO, lifting MATH accuracy from **56.4%** to **60.4%** (+4.0 points).

RESEARCH AND PUBLICATIONS

Phase-Adaptive Generation: Efficient Decoding for Diffusion Language Models [GitHub]

Feb 2026 – May 2026

- NYU project under Prof. Greg Durrett. Analyzed generation dynamics of Dream/LLaDA-style diffusion LMs via per-token refinement traces, revealing interpretable phase structure: template spans stabilize immediately while arithmetic and answer-finalization spans demand substantially more refinement.
- Trained Transformer predictors over 138K block-level control tuples to forecast per-block refinement budgets; integrated into the LLaDA decoding loop, matching AdaBlock's **89.5% GSM8K accuracy** with approximately **21% fewer forward evaluations**. [Report]

ML-Driven Program Phase Prediction for Dynamic Hardware Resource Optimization [GitHub]

Jan 2026 – May 2026

- Independent study with Prof. Mohamed Zahran, NYU Courant. Clustered temporal traces of multicore workloads into interpretable pressure states and distilled them into compact predictors with sub-KB model state, reaching **90%+** phase-prediction accuracy while keeping per-timestep overhead negligible.

Publications

2024 – 2025

- *Advancements in Cache Management: A Review of Machine Learning Innovations* — Frontiers in AI, 2025. Single-author, 10+ citations; reviewed 40+ papers on ML for cache management.
- *LLC Intra-set Write Balancing for Non-Volatile Memory Caches* — arXiv, 2024. Bachelor's thesis research.

TECHNICAL SKILLS

Product / full-stack: React, JavaScript, HTML/CSS, FastAPI, Postgres, Redis, Docker, Kubernetes, Git, Linux, developer enablement and technical documentation

LLMs and GenAI: foundation model integration and deployment, production-ready GenAI capabilities, scalable service integration, Model Ops (model hubs, versioning, checkpointing), agentic systems and multi-agent orchestration, tool-use orchestration, reasoning pipelines, agent evaluation frameworks, evaluation and benchmarking pipelines, continuous evaluation and automated model testing, retrieval-augmented generation, vector search (Qdrant), context management, compound AI systems, SFT/GRPO/RLHF, PyTorch, Hugging Face, tokenizers, vLLM, inference optimization, Anthropic and OpenAI APIs, MCP

Data / infrastructure: Python, C/C++, Spark, Hive, Airflow, Triton kernel development, NVIDIA Nsight Systems, Slurm/HPC, cloud AI infrastructure (GCP, Azure), reproducible pipelines